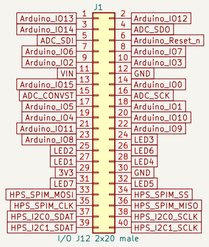
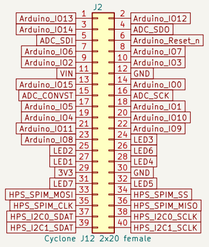
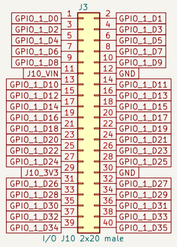
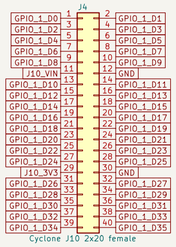
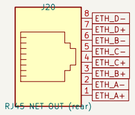
J12 and J10 (GPIO_1) are passed through pin for pin. Net names are the base board names. J10 pins 11/29 are named J10_VIN/J10_3V3: same supply as VIN/3V3, tied on both QMTECH boards, not linked on the backplane. (QMTECH schematic 2024-06-07, page 5).
J12 pins 12 and 30 are GND, pin 32 is LED5 (the A4 table was wrong).

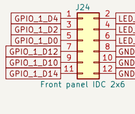


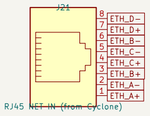




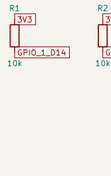




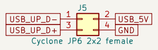


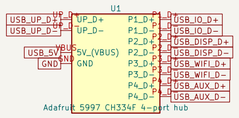


Network extension: J21 gets a short patch cable from the Cyclone RJ45 (P1, front). J20 is the rear panel jack. Pin N to pin N, no magnetics (the Cyclone RJ45 has them).

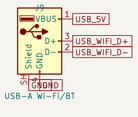


USB: Cyclone JP6 (USB3300 host) -> Adafruit 5997 USBHOST. JP6 shares D+/D− with the Cyclone J13 USB-A: leave J13 empty.
Hub power = JP6 pin 2 (USB_5V, current limited by the Cyclone U13 TPS2553/TPS22211). It also feeds the I/O USB 5V,
the AUX and DISPLAY headers and the Wi-Fi USB-A. On the module all 5V pads and all GND pads are one net.
Hub ports: P1 = I/O Board, P2 = DISPLAY, P3 = Wi-Fi USB-A, P4 = AUX (P3/P4 swapped against the brief to avoid vias).



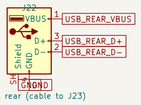






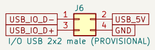


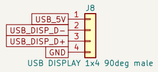




Front panel (J24, pairs from the west): RESET (J10.17), OSD (J10.13), USER (J10.15): signal + GND.
10k pull-up to 3V3, button to GND. LED USER (J10.1), LED DISK (J10.3), LED POWER (J10.5):
3V3 – 1k – LED – FPGA pin. The I/O Board may already have pull-ups/LEDs on these pins (in parallel).

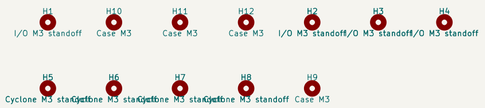








J6 is PROVISIONAL: it assumes the I/O Board V2.1 has a 2x2 socket over the base JP6.
Check on the real board before ordering.



Generated by tools/build_sch.py from tools/design.py
I/O Board V2.1 left (upright), Cyclone V SoC right (flipped), Adafruit 5997 hub
PROTOTYPE – NOT VALIDATED – DO NOT FABRICATE

Sheet: /
File: MisterFlat_RevA5.kicad_sch

Title: QMTECH MiSTer flat backplane

Size: A2	Date: 2026-10-09	**Rev: A5**
KiCad E.D.A. 10.0.7		Id: 1/1